

Project Proposal for ByteMasters: Development of Explainable AI (XAI) based model for prediction of heavy /high impact rain events using satellite data

Proposed Solution:

Developing a nowcasting system for heavy precipitation rainfall events with an AI-based model, understanding key predictors, and incorporating explainability is a complex project that involves multiple steps and considerations.

Data Collection and Preprocessing:

- Collect historical weather data, including satellite data from INSAT-3D/3DR, The dataset regarding them can be found in MOSDAC, CMR Search.
- INSAT-3D is equipped with several instruments, including an imager and a sounder
 - *Imager*: This instrument captures visible and infrared images of the Earth's surface and atmosphere. These images are used to monitor cloud cover, weather systems, and various meteorological parameters.
 - *Sounder*: The sounder measures vertical temperature and humidity profiles in the Earth's atmosphere, which are essential for weather prediction.

Feature Selection and Engineering:

- Identify the most important predictors for heavy precipitation events, which may include variables like temperature, humidity, wind patterns, cloud cover, and historical rainfall data.
- Engineer relevant features from the collected data.

Model Development:

- Develop a machine learning or deep learning algorithm suitable for predicting heavy precipitation events. Some common choices are Convolutional Neural Networks (CNNs) and LSTM.

Explainable AI (XAI) Integration:

- Implement XAI techniques to make the model's predictions interpretable. This may involve techniques like LIME (Local Interpretable Model-agnostic Explanations) or SHAP (Shapley Additive explanations).
- Provide explanations for model predictions, helping users understand why a certain prediction was made.

Web Application Development:

- Create a web application that allows users to input parameters (e.g., date, time, location) and receive predictions for heavy precipitation events.
- Display model predictions and explanations in a user-friendly manner.

Evaluate and Monitor:

- Evaluate the model's accuracy using appropriate metrics, design an inference pipeline, use MLOP tools to monitor the model's real-time performance and update the model to adapt to changing weather patterns and improve accuracy.
- Keep track of previous weather predictions based on satellite information of a particular location in the dashboard.

Integration with INSAT-3D/3DR Data:

- Ensure that the model can ingest real-time or near-real-time satellite data for current weather conditions.
- Implement mechanisms for data synchronization and automatic updates.

User Education and Training:

- Provide documentation and training for users of the web application, explaining how to interpret model predictions and understand the XAI component.

Deployment and Maintenance:

- Deploy the web application on a scalable and reliable server or cloud platform.
- Continuously maintain and update the system to adapt to changing conditions and improve accuracy.

Failures and Error Handling:

- Understand that models can fail for various reasons, such as sudden weather pattern changes, data inaccuracies, or limitations in the model itself.
- Implement error handling and reporting mechanisms to alert users when the model's predictions may be less reliable.

References:

- <https://www.sciencedirect.com/science/article/abs/pii/S2352938517301544>
- https://www.mosdac.gov.in/docs/INSAT3D_Products.pdf
- <https://www.tropmet.res.in/ip4/ppt/28112019/2%20CMK-FINAL-IITM.pdf>